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Sub ject : DSA

Unit 3, S1, SLO2 - Stack Implementation Using Array

Push Operation - Algorithm to insert an element in a stack
Step 1: IF TOP = MAX-1
PRINT _____________
Goto Step 4
[END OF IF]
Step 2: SET TOP = ___________
Step 3: SET STACK[______] = VALUE
Step 4: END

Answer: Step 1: PRINT OVERFLOW
Step 2: SET TOP = TOP + 1
Step 3: SET STACK[TOP] = VALUE
Step 4: END

Pop Operation - Algorithm to delete an element from a stack
Step 1: IF TOP = _____
PRINT UNDERFLOW
Goto Step 4
[END OF IF]
Step 2: SET VAL = STACK[______]
Step 3: SET TOP = _________
Step 4: END

Answer: Step 1: IF TOP = -1
Step 2: SET VAL = STACK[TOP]
Step 3: SET TOP = TOP - 1
Step 4: END

Peek Operation - Algorithm for Peek operation
Step 1: IF TOP = _______
PRINT STACK IS EMPTY
Goto Step 3
[END OF IF]
Step 2: RETURN STACK[_______]
Step 3: END

Answer: Step 1: IF TOP = -1
Step 2: RETURN STACK[TOP]
Step 3: END